IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<09/01/2026>

Executive Summary

- This project analyzes historical SpaceX launch data to identify factors influencing mission success.
- Exploratory Data Analysis (EDA), SQL analysis, interactive maps, and dashboards were applied.
- Key variables analyzed include payload mass, orbit type, launch site, and flight experience.
- Several machine learning classifiers were trained using standardized data and GridSearchCV.
- The best-performing model achieved high prediction accuracy and reliable classification results.
- Insights support data-driven decision making for future mission planning.

Introduction

- SpaceX focuses on reducing launch costs by reusing rocket boosters.
- Successful landings are critical for operational efficiency and cost reduction.
- Predicting launch success helps optimize mission design, safety, and resource allocation.
- This project investigates historical launch data to discover trends and build predictive models

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection – SpaceX API

- Data was collected using REST API calls from the SpaceX public API. The main launches endpoint was queried to retrieve launch records and extract related object IDs. Additional API calls were performed to obtain rocket details, payload characteristics, and launch site information. All datasets were merged and cleaned before exporting the final structured dataset for analysis and machine learning.
- The Github URL of the notebook is: https://github.com/ivan-codez/Capstone-Proyect/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

```
[Start]
   ↓
[GET Launches]
   ↓
[Extract IDs]
   ↓
[GET Rockets]
   ↓
[GET Payloads]
   ↓
[GET Launchpads]
   ↓
[Merge DataFrames]
   ↓
[Clean + Engineer Features]
   ↓
[Export CSV]
   ↓
[End]
```

Data Collection - Scraping

- Web scraping was used to collect launch-related information from an online HTML source. The target web page was accessed using HTTP requests, and the HTML content was parsed using BeautifulSoup. Relevant tables and tags were identified and extracted, then cleaned and structured into a pandas DataFrame. The processed data was finally exported to a CSV file for further analysis and integration with the SpaceX API dataset.
- The Github URL is: https://github.com/ivan-codez/Capstone-Proyect/blob/main/jupyter-labs-webscraping.ipynb

```
START
  ↓
Define Target URL (Wikipedia / HTML page)
  ↓
Send HTTP Request (requests.get)
  ↓
Receive HTML Response
  ↓
Parse HTML with BeautifulSoup
  ↓
Locate Tables / Tags (find, find_all)
  ↓
Extract Relevant Fields
  ↓
Clean and Format Data
  ↓
Convert to Pandas DataFrame
  ↓
Save to CSV
  ↓
END
```

Data Wrangling

- Raw data collected from the SpaceX API and web scraping sources were merged and cleaned to build a structured dataset. Irrelevant columns were removed, missing values were handled, and text fields were standardized. A binary target label was created from mission outcomes. Categorical features were transformed using one-hot encoding and all variables were cast to numeric format. The final cleaned dataset was exported as a CSV file for exploratory analysis and machine learning modeling.
- The Github URL is: https://github.com/ivan-codez/Capstone-Proyect/blob/main/labs-jupyter-spacex-data%20wrangling_jupyterlite.ipynb

```
START
   ↓
Load Raw Data (API + Web Scraping CSVs)
   ↓
Merge Datasets (join by IDs / keys)
   ↓
Select Relevant Columns
   ↓
Handle Missing Values
   ↓
Clean Text Fields (strip, rename, normalize)
   ↓
Create Target Label (Class / Landing_Class)
   ↓
Encode Categorical Variables (One-Hot Encoding)
   ↓
Cast Data Types (float64)
   ↓
Export Clean Dataset to CSV
   ↓
END
```

EDA with SQL

- What was done in the notebook was the following:
- Counted total launches and outcomes.
- Computed success rate by launch site.
- Computed success rate by orbit type.
- Analyzed payload distributions.
- Measured launch frequency.
- Filtered and summarized Special cases.
- The Github URL: https://github.com/ivan-codez/Capstone-Proyect/blob/main/Complete%20EDA%20with%20SQL.ipynb

EDA with Data Visualization

- Multiple visualizations were created to explore relationships between mission success and key operational variables. Scatter plots were used to analyze payload mass versus orbit type and flight number versus launch site. Bar charts were applied to compare success rates across orbit categories. Line charts were used to observe yearly success trends over time. These visualizations helped identify patterns, correlations and operational trends that support data-driven insights.
- The Github URL is: https://github.com/ivan-codez/Capstone-Proyect/blob/main/EDA%20with%20Data%20Visualization.ipynb

Predictive Analysis (Classification)

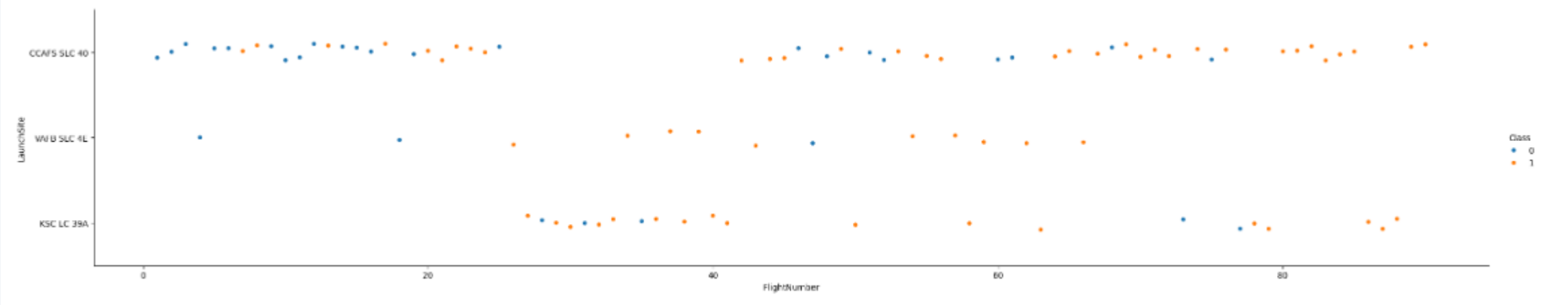
- Multiple classification models were trained to predict launch success, including Logistic Regression, Support Vector Machine, Decision Tree and K-Nearest Neighbors. The dataset was standardized and split into training and testing sets. Hyperparameters were optimized using GridSearchCV with cross-validation. Models were evaluated using accuracy metrics and confusion matrices. The best-performing model was selected based on test performance and generalization capability.
- The Github URL is: https://github.com/ivan-codez/Capstone-Proyect/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

```
START
  ↓
Load Clean Dataset
  ↓
Split Data (Train / Test)
  ↓
Standardize Features
  ↓
Train Models
(Logistic, SVM, Decision Tree, KNN)
  ↓
Hyperparameter Tuning (GridSearchCV)
  ↓
Cross Validation
  ↓
Evaluate Models (Accuracy, Confusion Matrix)
  ↓
Compare Results
  ↓
Select Best Model
  ↓
END
```

Section 2

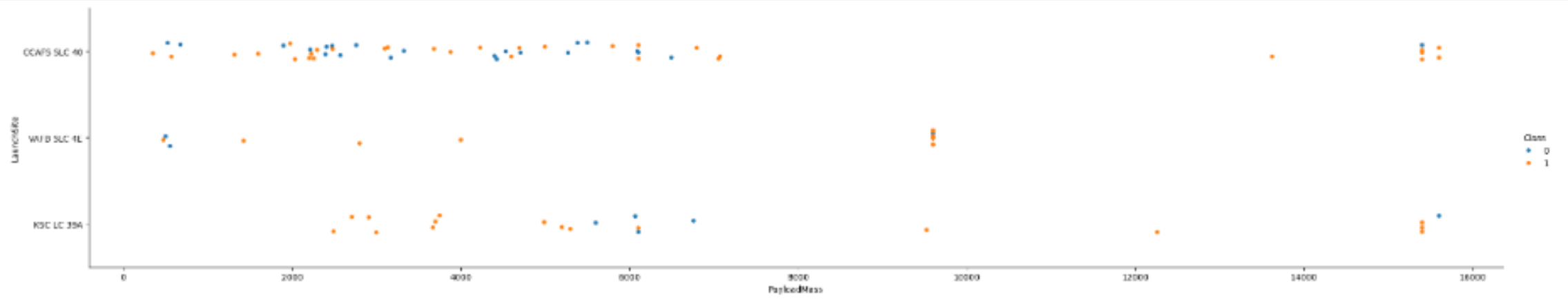
Insights drawn from EDA

Flight Number vs. Launch Site



- The scatter plot shows the relationship between flight number and launch site, with color indicating mission success. As the number of flights increases, successful launches become more frequent, indicating operational learning and improved reliability over time. Differences in success patterns can also be observed between launch sites.

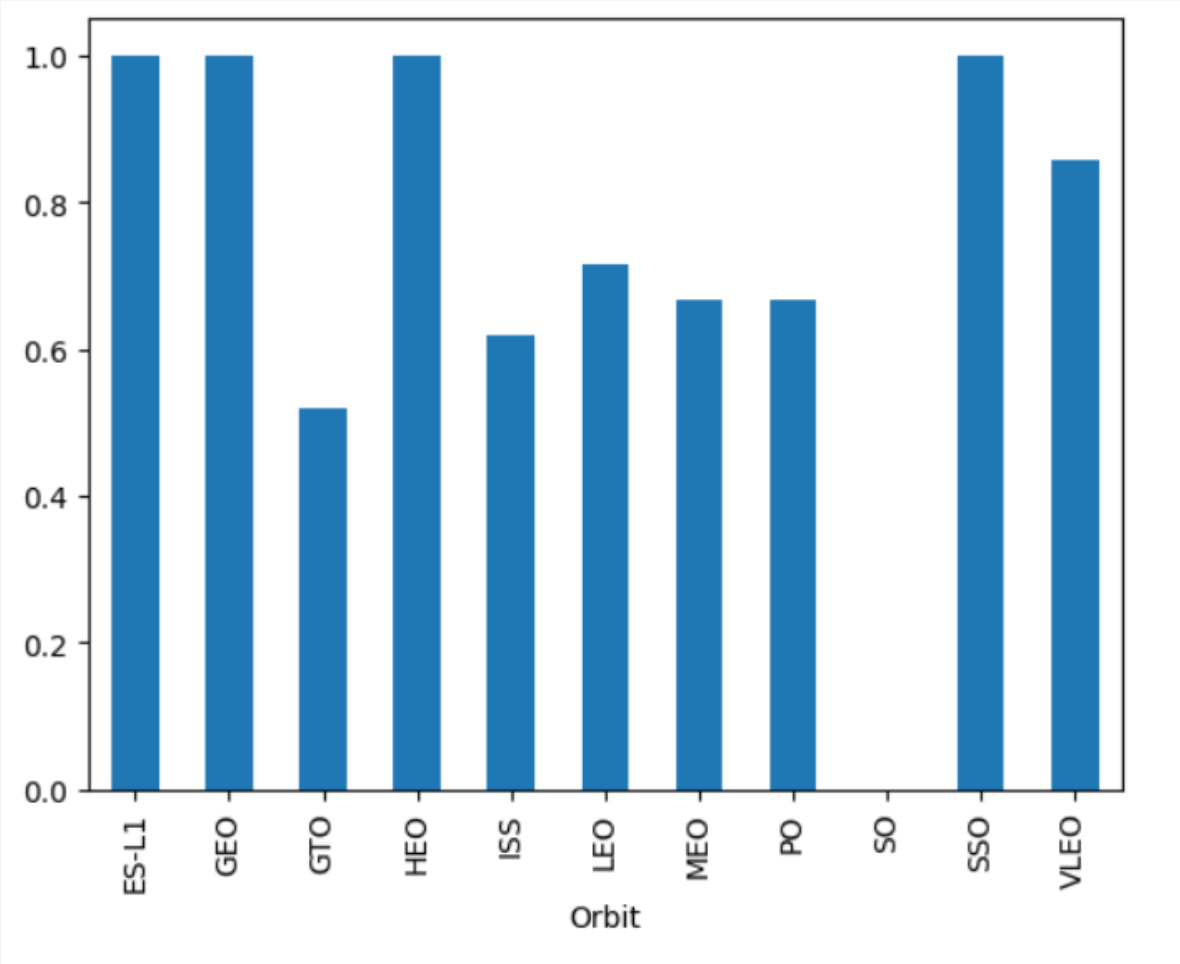
Payload vs. Launch Site



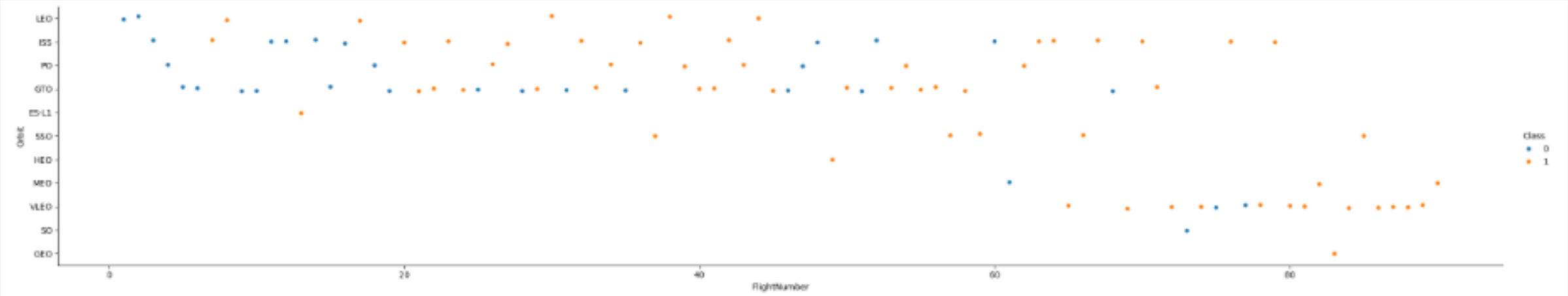
- The scatter plot illustrates the relationship between payload mass and launch site, with color indicating mission success. It can be observed that VAFB SLC-4E does not handle heavy payloads above approximately 10,000 kg, while CCAFS SLC-40 and KSC LC-39A support a wider range of payload masses, including heavier missions. Successful launches are more frequent at higher payload ranges for CCAFS and KSC, whereas VAFB mainly operates in lower payload regimes. This suggests that payload capacity and operational capability vary by launch site and influence mission outcomes.

Success Rate vs. Orbit Type

- The bar chart shows the success rate across different orbit types. Orbits such as ES-L1, GEO, HEO, and SSO exhibit near-perfect success rates, indicating high operational reliability for these mission profiles. VLEO also demonstrates a high success rate, while LEO, MEO, and PO present moderate success levels. In contrast, GTO shows a noticeably lower success rate compared to other orbits, reflecting the higher technical complexity and operational challenges associated with transfer orbits. Overall, the results suggest that mission success strongly depends on orbit type, with more demanding orbital missions exhibiting greater variability in outcomes.

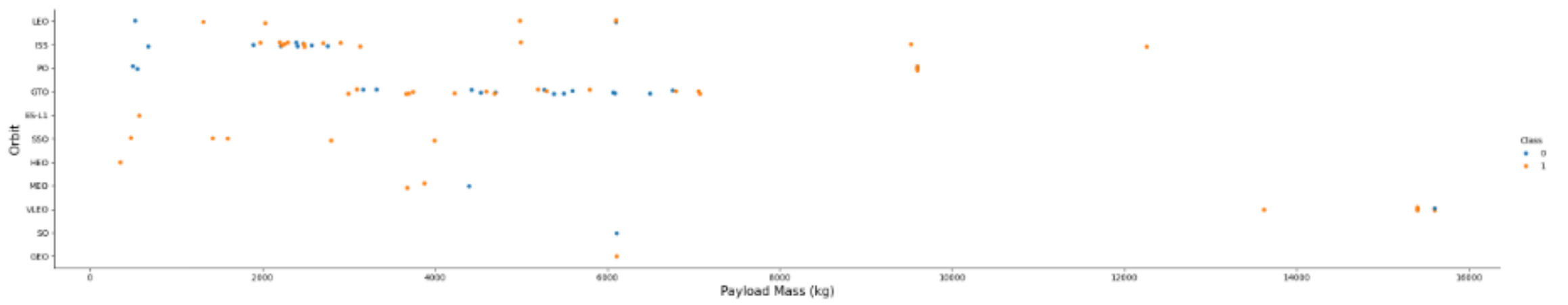


Flight Number vs. Orbit Type



- The scatter plot illustrates the relationship between flight number and orbit type, with color indicating mission success. For LEO missions, a clear increase in successful outcomes is observed as the flight number increases, reflecting improved operational reliability over time. Other orbit types such as ISS and PO show a mix of successful and unsuccessful launches, while GTO displays greater variability with no strong dependency on flight number. Overall, the visualization suggests that accumulated flight experience positively influences mission success for certain orbit categories, particularly LEO, whereas more complex orbits maintain higher uncertainty regardless of experience.

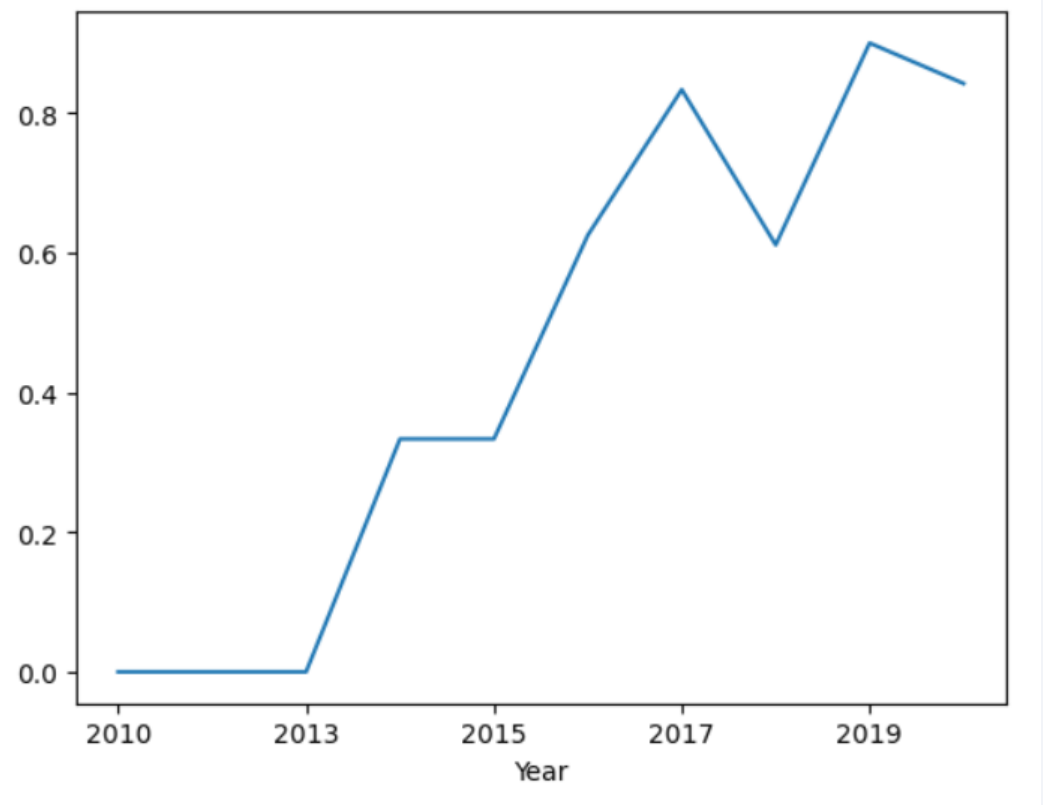
Payload vs. Orbit Type



- The visual analysis shows that launch performance depends on both operational experience and mission characteristics. Higher payload missions are mainly supported by CCAFS SLC-40 and KSC LC-39A, while VAFB SLC-4E operates only in lower payload ranges. Orbits such as ES-L1, GEO, HEO, and SSO exhibit consistently high success rates, whereas GTO presents lower reliability due to greater mission complexity. Additionally, LEO missions demonstrate increasing success as flight number grows, indicating learning effects, while payload-dependent success is more stable for Polar, LEO, and ISS orbits and more variable for GTO.

Launch Success Yearly Trend

- The yearly success trend shows a clear improvement in launch reliability over time, with near-zero success rates in the early years followed by a steady increase after 2013. Success rates rise significantly between 2014 and 2017, indicating rapid operational learning and technological maturation, with a slight dip around 2018 and a peak close to 2019–2020. Overall, the trend reflects continuous performance improvement and increasing mission stability as experience and system reliability grow.



All Launch Site Names

- The dataset contains three main launch sites: CCAFS LC-40 (Cape Canaveral Space Launch Complex 40), VAFB SLC-4E (Vandenberg Air Force Base Space Launch Complex 4E), and KSC LC-39A (Kennedy Space Center Launch Complex 39A). These sites differ in geographic location and operational capacity, which influences payload range, orbit accessibility, and overall launch performance.

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Launch Site Names Begin with 'KSC'

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_O
0	2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	
1	2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	
2	2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	
3	2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	
4	2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	

- This are 5 recorded Launch site names beginning with KSC corresponding to KSC LC-39A (Kennedy Space Center Launch Complex 39A).

Total Payload Mass

- The total payload mass in the dataset is 48,213 kg, representing the cumulative payload delivered across the selected launches.

	Total_Payload_Mass
0	48213

Average Payload Mass by F9 v1.1

- The average payload mass is approximately 2,928.4 kg, indicating the typical payload size carried per launch in the dataset.

	Avg_Payload_Mass
0	2928.4

First Successful Ground Landing Date

- The first successful drone landing occurred on April 8, 2016, marking a major milestone in SpaceX's reusable rocket operations.

	First_Successful_Drone_Landing
0	2016-04-08

Successful Drone Ship Landing with Payload between 4000 and 6000

- The boosters that successfully landed on a drone ship with payload mass between 4,000 kg and 6,000 kg are:
- F9 FT B1032.1, F9 B4 B1040.1, and F9 B4 B1043.1.

	Booster_Version
0	F9 FT B1032.1
1	F9 B4 B1040.1
2	F9 B4 B1043.1

Total Number of Successful and Failure Mission Outcomes

- The table shows the distribution of mission outcomes in the dataset: there is 1 failure in flight, 98 successful missions, 1 success with unclear payload status, and 1 additional success record, indicating an overall very high mission success rate.

	Mission_Outcome	Total
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

Boosters Carried Maximum Payload

- The boosters that carried the maximum payload are:
- F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, and F9 B5 B1049.7.
- These boosters were associated with the highest payload mass values in the dataset.

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2015 Launch Records

- The table shows successful ground pad landings by month, including the booster version and launch site. Most of these successful landings occurred at KSC LC-39A, with one occurrence at CCAFS SLC-40, indicating that KSC LC-39A was the primary site for ground pad recovery during the observed period.

	Month	Landing_Outcome	Booster_Version	Launch_Site
0	02	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
1	05	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2	06	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
3	08	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
4	09	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
5	12	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The table summarizes landing outcomes across missions: 10 launches had no landing attempt, 5 were successful drone ship landings, 5 failed drone ship landings, and 3 successful ground pad landings. Additionally, there were 3 controlled ocean landings, 2 uncontrolled ocean landings, 2 parachute failures, and 1 precluded drone ship landing, reflecting the diversity of recovery strategies and outcomes during different mission phases.

	Landing_Outcome	Total
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

Section 3

Predictive Analysis (Classification)

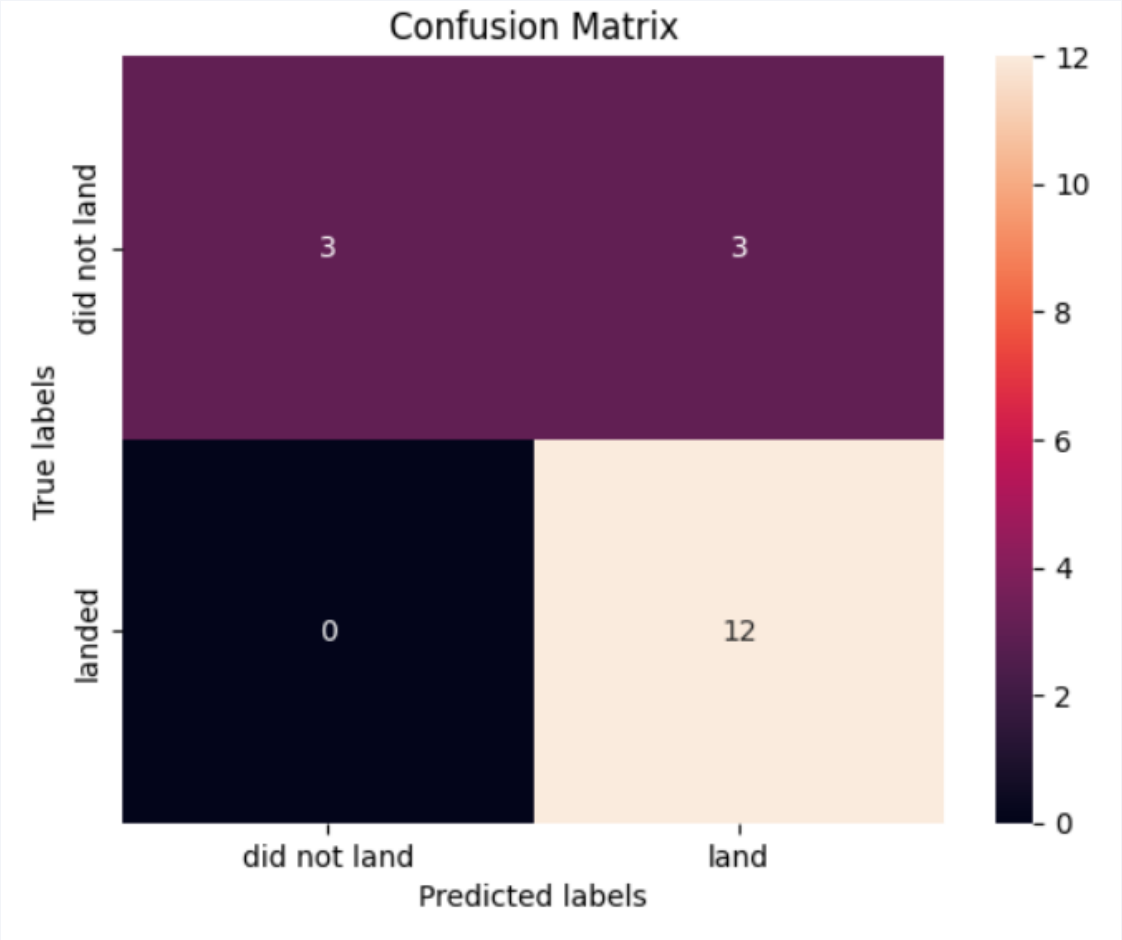
Classification Accuracy

```
LogReg test accuracy: 0.8333333333333334
SVM test accuracy: 0.833333333333334
DecisionTree test accuracy: 0.7777777777777778
KNN test accuracy: 0.833333333333334
```

- The test results show that Logistic Regression, SVM, and KNN achieved the highest accuracy of approximately 83.3%, while the Decision Tree model reached a slightly lower accuracy of approximately 77.8%. This indicates that linear and distance-based models provided better generalization performance on the test dataset compared to the tree-based approach.

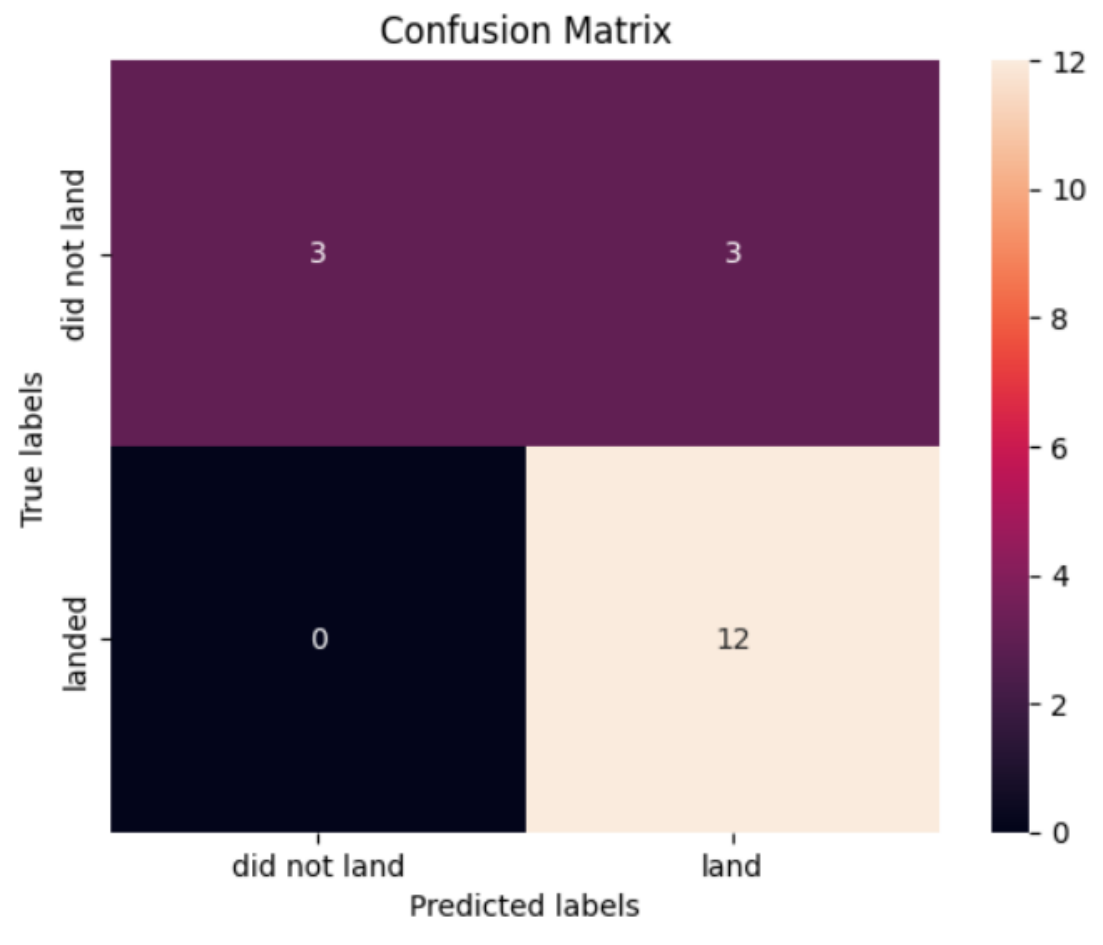
Confusion Matrix - KNN

- The KNN confusion matrix shows 12 true positives (successful landings correctly predicted), 3 true negatives (failed landings correctly predicted), 3 false positives (failed landings predicted as successful), and 0 false negatives (successful landings missed). This indicates that the model is very effective at identifying successful landings, but occasionally overestimates success by producing some false positives.



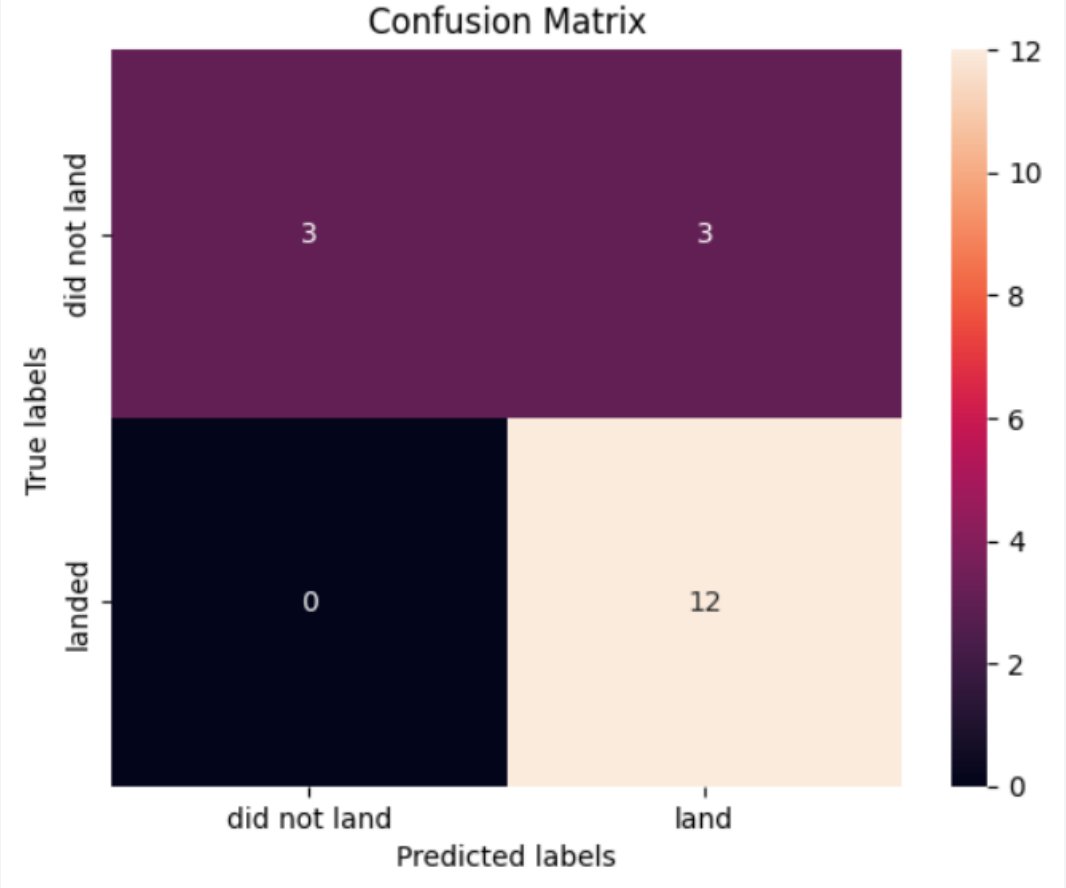
Confusion Matrix - SVM

- The confusion matrix indicates that the model correctly identified 12 successful landings and 3 failed landings, with no false negatives, meaning all actual successful landings were correctly predicted. However, there are 3 false positives, where failed landings were incorrectly classified as successful. Overall, the model demonstrates strong sensitivity in detecting successful landings, with minor overestimation of success.



Confusion Matrix - LOGREG

- The confusion matrix shows that the model correctly classified 12 successful landings and 3 failed landings, with no false negatives, meaning all actual successful landings were accurately detected. However, there are 3 false positives, where failed landings were predicted as successful. This indicates high sensitivity and strong overall performance, with a slight tendency to overestimate successful outcomes.



Conclusions

- Launch performance improves with operational experience, showing higher success rates as flight numbers increase, especially for LEO missions.
- Payload capacity varies by launch site, with CCAFS SLC-40 and KSC LC-39A supporting heavier payloads than VAFB SLC-4E.
- Orbit type strongly influences mission success, with ES-L1, GEO, HEO, and SSO showing the highest reliability, while GTO presents greater variability and lower success rates.
- Spatial analysis confirms that launch sites are strategically located near coastlines and close to the equator to optimize launch efficiency and safety.
- Predictive models achieved strong performance, with Logistic Regression, SVM, and KNN reaching approximately 83% accuracy.
- Data-driven analysis demonstrates that payload mass, orbit type, and operational maturity are key drivers of launch success.

Thank you!